# WaterBuddy planning and design

Screen designs and app logic for the WaterBuddy Python project

## THE USER PROBLEM

It is easy to forget to drink water during a busy school or work day. WaterBuddy keeps tracking simple: choose a goal, log a drink and see how much is left.

### Quick to use

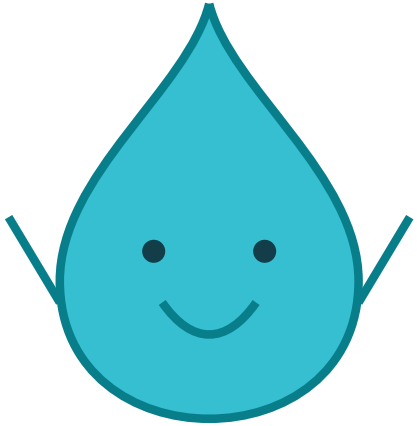
A visible +250 ml button, optional container fractions and a custom field reduce repeated typing.

### A little encouragement

A friendly drop reacts at 50%, 75% and 100%. Points stop increasing after completion; there is no drinking leaderboard.

### No account needed

No account or personal identifier. Records exist in the current server session; downloads preserve a readable personal record.



## DESIGN OBJECTIVE

Choose a goal, log actual water, understand progress and build a manageable routine.

### Research approach

The design uses the assignment brief and published health guidance. The user problems are starting assumptions for the design; they have not been tested through interviews or a survey.

# Age aware goals and first use

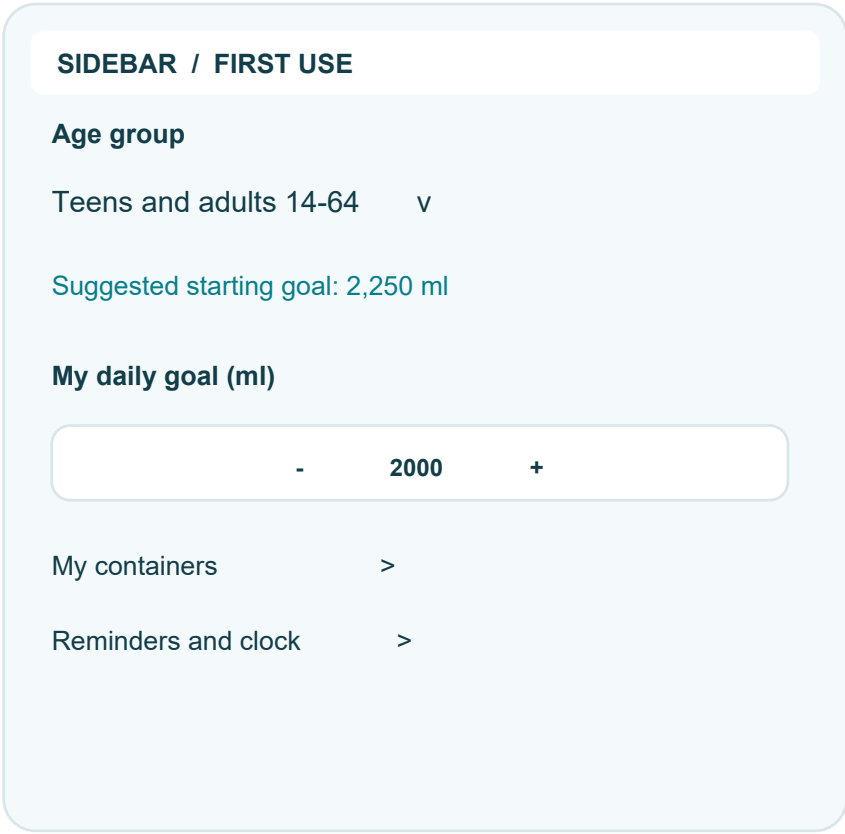
Settings remain in the sidebar so users can change their goal without losing water already logged

## ILLUSTRATIVE LOGGING PRESETS

| Age group | Starting goal |
|---|---|
| Children 4-8 | 1,200 ml |
| Young people 9-13 | 1,700 ml |
| Teens and adults 14-64 | 2,250 ml |
| Older adults 65+ | 1,800 ml |

These are editable classroom presets carried forward from the project concept. They are not universal drinking-water recommendations. The app labels them as starting goals.

The About page separately explains EFSA total-water references, which include food and all drinks. A user logging plain water should not confuse these two quantities. [1]




Action: select an age group. Python looks up its preset. Adjust the goal from 250 to 5,000 ml; existing logs stay intact.

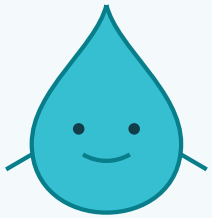
### Age-aware support

Children choose an appropriate goal with an adult. Individual clinical guidance takes priority over any app preset.

# Home and milestone reactions

Four states of the same dashboard | Illustration uses an adjusted 2,000 ml goal

**HOME / 0%**



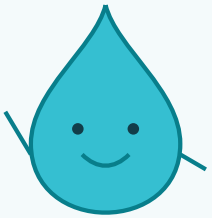
**0 / 2,000 ml**

2,000 ml remaining

Ready when you are

No logs yet. Total = 0; progress = 0%; buddy is ready.

**HOME / 50%**



**1,000 / 2,000 ml**

1,000 ml remaining

Halfway there

At 50%, the drop smiles and waves. Remaining = 1,000 ml.

**HOME / 75%**



**1,500 / 2,000 ml**

500 ml remaining

Your buddy is cheering

At 75%, both arms rise. The message changes immediately.

**HOME / 100%**



**2,000 / 2,000 ml**

0 ml remaining

Goal complete

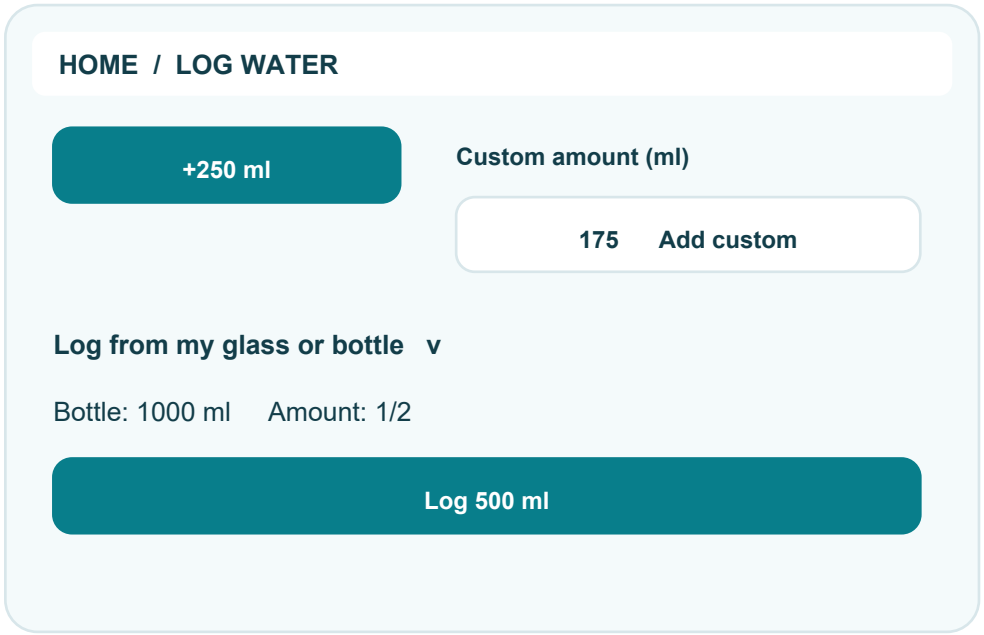
At 100%, celebrate and unlock Hydration Hero. Daily rewards are capped.

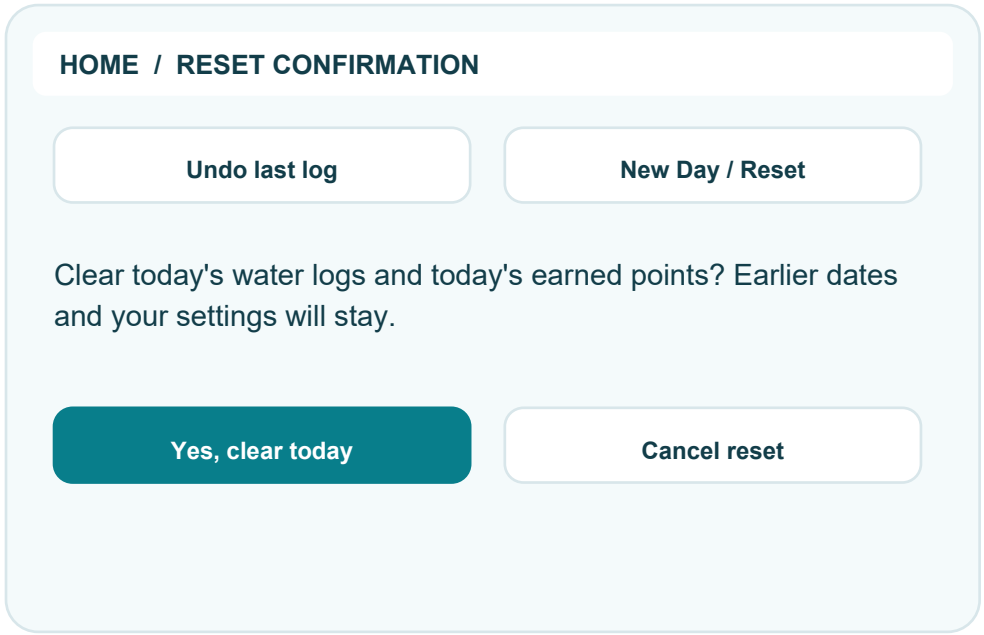
## ONE CONSISTENT DASHBOARD

Home shows total, remaining, chosen goal, true percentage, progress bar and a comparison with the age preset. At 2,250 / 2,000 ml, show 112.5%, remaining 0 ml and a full bar.

# Logging and a safe reset

Controls are sections within Home, matching the final app rather than separate setup screens




1 Quick log appends 250 ml. Custom input accepts whole values from 1 to 2,000 ml. Invalid values cannot enter the log.

2 Container estimate = size x fraction. A 1,000 ml bottle x 1/2 adds 500 ml. Fractions are 1/4, 1/2, 3/4 and full.

3 Undo removes only the last drink. All totals, visuals and points recalculate from the remaining log.

4 Cancel keeps the data. Confirm clears today only. A reset does not create a new date or increase a streak.

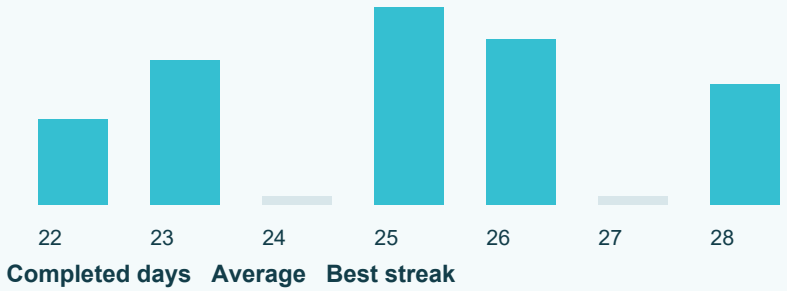
### Calendar rollover

The next actual local date starts a fresh record automatically on interaction, or on the 60 second Home check. Earlier session records stay available.

# Progress and rewards

Example layouts for weekly records and daily rewards

**PROGRESS / LAST SEVEN DAYS**




**Completed days** **Average** **Best streak**

**Download weekly CSV** **Session backup JSON**

**REWARDS / DAILY ACHIEVEMENTS**

| AquaPoints | Calendar streak |
|---|---|
| Up to 125 today | Consecutive real dates |

50% Halfway Hero

75% Steady Sipper

100% Hydration Hero

At 100% choose a celebration accessory

A dated chart, status table and today's drink log explain the numbers. Missing dates say No record; the average uses recorded dates only.

Points = min(100, floor(total / goal x 100)) + 25 on completion. A total above the goal earns no extra points.

History belongs to this session. JSON and CSV are readable exports; automatic backup restore and permanent history are outside this version.

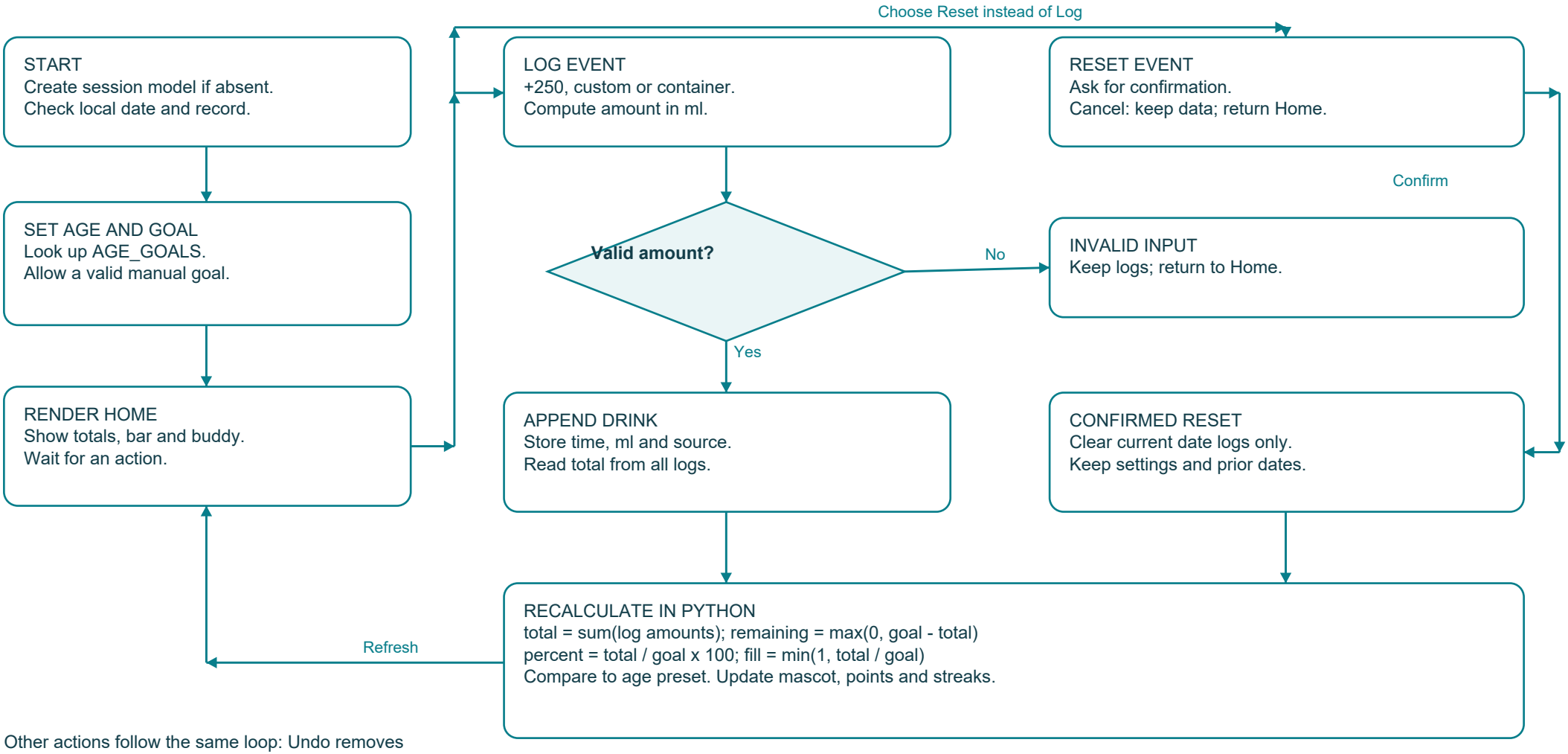
A streak counts saved, consecutive calendar dates meeting each day's goal. Goal changes, undo and reset recalculate points and badges.

### Optional features support the core task

Mini missions use 25%, 50%, 75% and 100% checkpoints. Celebration accessories are cosmetic; there are no promised notifications or rewards outside the app.

# Main Python logic and event flow

Follow the arrows from initialization to an action, then back to the refreshed interface




Other actions follow the same loop: Undo removes the final log; age or goal changes update the goal and preserve intake. [Python]

# Feedback decisions and reminders

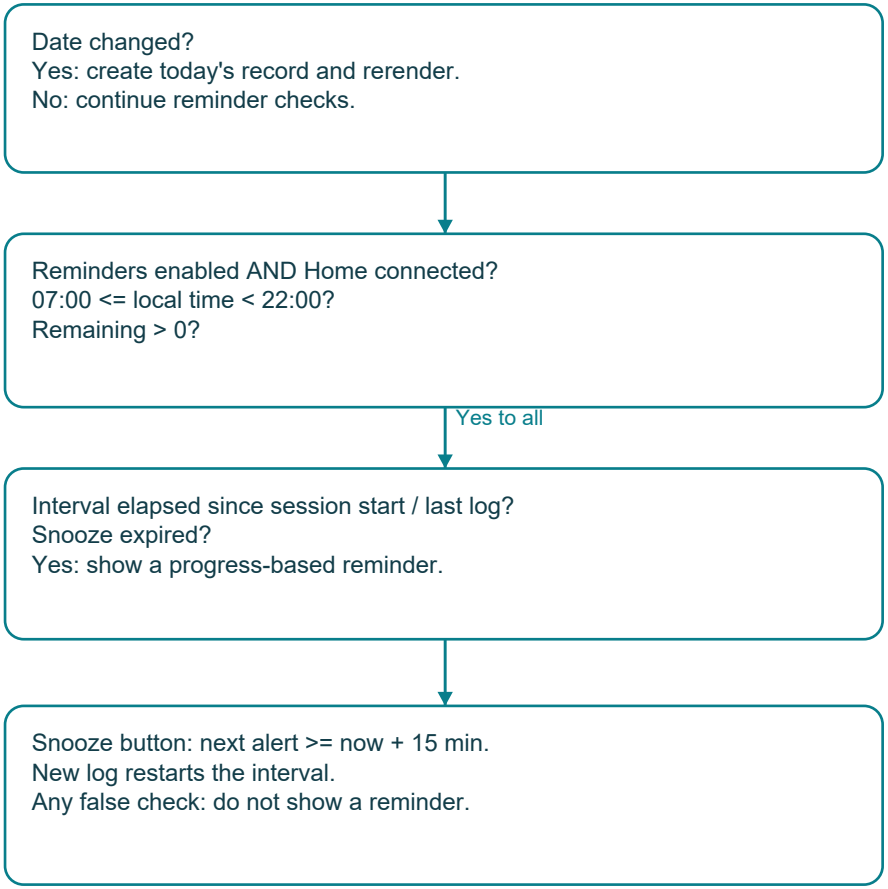
Detailed rules prevent stale totals, duplicate rewards and misleading notifications

## MILESTONE PRIORITY

| Check in this order | Buddy reaction | Output |
|---|---|---|
| percent >= 100 | Complete | Goal complete and full bar |
| else percent >= 75 | Cheer | 75% encouragement |
| else percent >= 50 | Wave | Halfway message |
| else percent > 0 | Smile | Good start |
| else | Ready | Log your first drink |

Check the highest threshold first, so a completed goal never shows a halfway message. Visual fill is capped but the numerical percentage is accurate above 100%.

## ON EACH 60 SECOND HOME CHECK




### State stored in st.session_state

model = age + chosen goal + current date + dated logs.
Each drink stores ml, time and source.
The interface uses Streamlit widgets; SVG replaces desktop Turtle so mascot reactions work in the deployed web app.

# From storyboard to the FA2 app

Every planned control has a direct implementation and a checkable outcome

| FA1 plan | FA2 location | Evidence to demonstrate |
|---|---|---|
| Age group and adjustable goal | Sidebar; age_changed / goal_changed | Change age; preset changes; logs remain |
| Quick log and containers | Home; log; container_amount | +250 adds 250; half a 1 L bottle adds 500 |
| Live calculations and visuals | current; progress; Home | 2,250 / 2,000 gives 112.5%, 0 remaining, full bar |
| Milestone mascot reactions | milestone; mascot_svg | Set 2,000 ml; reach 1,000, 1,500 and 2,000 |
| Confirmed New Day reset | Home; reset_today | Cancel keeps data; confirm clears today |
| History and motivation | Progress and Rewards | Real-date streaks; capped points; exports |
| Optional reminders | reminder_panel on Home | On-screen only; quiet hours and snooze |

## FA1 RUBRIC COVERAGE

Research and problem understanding 5 marks: user barriers, age-aware goals and annotated design decisions.
App flow and logic planning 10 marks: event loop, amount validation, formulas, reset, date rollover and milestone priority.
Visual storyboard 5 marks: settings, Home states, logging, confirmation, Progress, Rewards and reminder behavior.

# Research notes and references

Primary sources support the health context and the choice of deployment platform

## HEALTH CONTEXT AND DESIGN IMPLICATIONS

Water supports normal body function and temperature regulation. Dehydration can affect thinking and mood. Keeping a reusable bottle visible and offering water with meals are practical habit cues. These findings support a visible log button and short tips rather than complex health claims. [2]

EFSA total-water reference values include food and all drinks: ages 4-8, 1,600 ml; ages 9-13, 1,900 ml for girls and 2,100 ml for boys; ages 14 and over, including older adults, 2,000 ml for females and 2,500 ml for males. The app explains this distinction rather than labelling its classroom presets as health standards. [1]

## SOURCES

**[1] EFSA Dietary Reference Values summary report**

https://www.efsa.europa.eu/sites/default/files/2017_09_DRVs_summary_report.pdf

**[2] CDC About Water and Healthier Drinks**

https://www.cdc.gov/healthy-weight-growth/water-healthy-drinks/

**[3] Streamlit Community Cloud deployment guide**

https://docs.streamlit.io/deploy/streamlit-community-cloud/deploy-your-app/deploy

**[4] Streamlit dependency management**

https://docs.streamlit.io/deploy/concepts/dependencies

Sources reviewed 28 October 2026. The supplied FA1 and FA2 briefs define the assessment requirements. The original WaterBuddy storyboard supplies the drop mascot and habit-building concept; this revision aligns the plan with the implemented app.